

Abdul Munim Baig

Computer Science Graduate | AI & Backend Systems Engineer

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Professional Summary

Computer Science graduate with expertise in machine learning, backend systems, and AI orchestration. Experienced in building enterprise-grade AI agents, distributed architectures, and automated workflows. Proficient in Python, C++, and Java, with a strong focus on solving complex engineering challenges.

Education

Bachelor of Science in Computer Science

FAST National University of Computer and Emerging Sciences, Islamabad

2022 – 2026

Experience

Software Engineer Intern, INZENT

Mar – Jul 2026

- Contributing to open-source PostgreSQL-based database tools and system-level debugging.
- Researching database internals to enhance performance and system reliability.

Intern — IT Department, Pakistan Tobacco Company

Jun – Jul 2025

- Designed and deployed a Leave Management System using Power Platform.
- Automated approval workflows, improving internal operational efficiency.

Machine Learning Engineer Intern, Genesys Research Lab

Jun – Aug 2024

- Preprocessed and analyzed large-scale datasets for ML model training.
- Developed backend services using Node.js and MongoDB.

Projects

AI Virtual Assistant (AIVA) — Final Year Project — *WordPress, Python, Flask, Groq*

- **Commerce Automation:** Built an AI-powered engine for WordPress to automate customer engagement, order handling, and payment verification across social platforms.
- **Multimodal Intelligence:** Integrated NLP, voice processing, and visual search using intent-driven state machines to detect buyer intent and scale live selling.
- **Hybrid LLM Orchestration:** Architected a dual-LLM system (Groq Llama 3.3 + Local 7B) to optimize for high-speed reasoning and conversational depth.
- **Semantic Search & RAG:** Leveraged a Vector Database (Sentence-Transformers) to implement intelligent product discovery and RAG-based query handling.

Multimodal RAG System for Document Analysis — *Python, FAISS, CLIP, LLaMA 3*

- Developed an end-to-end RAG pipeline for complex PDFs combining text and image embeddings.

Distributed File System (Mini-IPFS) — *C++, SHA-1, B-Tree*

- Implemented a ring-based DHT architecture with B-Tree indexing for fault tolerance.

Technical Skills

Languages: Python, Java, C++, JavaScript, SQL

Web & Frameworks: React, Spring Boot, Node.js, Flask, Docker, Google Cloud Run

AI & Data: TensorFlow, PyTorch, NumPy, Sentence-Transformers, FAISS

Databases: MySQL, MongoDB, Vector Databases (ChromaDB/FAISS)